

JEE Main - 7 | JEE 2022 (Gen 1 & 2)

Date: 01/02/2021

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

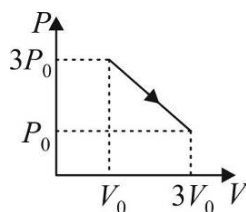
OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

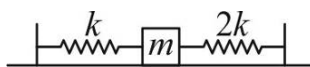
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A simple pendulum consists of a metal wire of coefficient of linear thermal expansion α attached to a heavy bob. The time period of this pendulum on a cold winter day with temperature 5°C is T_C , and its time period on a hot summer day with temperature 35°C is T_H . The ratio $\frac{T_C}{T_H}$ is close to:
 (A) $1+30\alpha$ (B) $1-30\alpha$ (C) $1+15\alpha$ (D) $1-15\alpha$
- One mole of a monoatomic gas at 27°C is allowed to expand isothermally such that its final volume is 8 times its initial volume. The work done by the gas during the expansion is: ($\log_e 2 = 0.693$)
 (A) $576 J$ (B) $1728 J$ (C) $5183 J$ (D) $466 J$
- A particle executes SHM along a straight line with frequency f and amplitude A . The minimum time taken by the particle to go from its mean position to a point at a distance $\frac{A}{2}$ from the mean position is:
 (A) $\frac{1}{6f}$ (B) $\frac{1}{8f}$ (C) $\frac{1}{12f}$ (D) $\frac{1}{16f}$
- Let us assume that the sun and another star A are perfect spheres and perfect black bodies. The radius of star A is four times the radius of the sun and it radiates heat energy at exactly the same rate as the sun. If the surface temperature of the sun is 6000 K , the surface temperature (in Kelvin) of the star A is:
 (A) 8484 (B) 4242 (C) 3000 (D) 2000
- A gas undergoes the thermodynamic process shown in the P-V diagram. The work done by the gas in this process is:

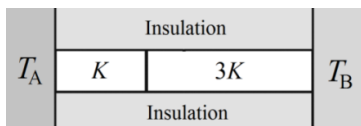


- (A) $3P_0V_0$ (B) $4P_0V_0$ (C) $\frac{9}{2}P_0V_0$ (D) $6P_0V_0$
- A small block of mass m is placed on a frictionless horizontal surface and connected to two massless springs of spring constant k and $2k$ as shown. The other ends of the springs are fixed to rigid walls. Initially, both springs are in their natural length. Now, the block is displaced by a small distance to the right and released. The time period of the resulting SHM is:

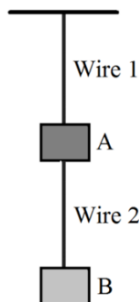


- (A) $2\pi\sqrt{\frac{m}{3k}}$ (B) $2\pi\sqrt{\frac{m}{2k}}$ (C) $2\pi\sqrt{\frac{2m}{3k}}$ (D) $2\pi\sqrt{\frac{3m}{2k}}$

7. Two cuboidal blocks made of different materials of thermal conductivity K and $3K$ are placed end to end between two heat reservoirs maintained at constant temperature T_A and T_B (such that $T_A > T_B$). The blocks have the same area of cross-section, but lengths L and $2L$ respectively. The temperature at the interface between the blocks is:



- (A) $T = \frac{1}{7}(T_A + 6T_B)$ (B) $T = \frac{1}{7}(6T_A + T_B)$
 (C) $T = \frac{1}{5}(2T_A + 3T_B)$ (D) $T = \frac{1}{5}(3T_A + 2T_B)$
8. Two blocks A and B of mass 4 kg and 2 kg respectively are suspended using two metallic wires as shown. The diameter of the wires are in the ratio $\frac{d_1}{d_2} = 1:2$, and their Young's moduli are in the ratio $\frac{Y_1}{Y_2} = 3:1$. The strain produced in them is in the ratio:



- (A) 2 : 1 (B) 4 : 9 (C) 2 : 3 (D) 4 : 1
9. A simple pendulum consists of a massless string of length L attached to a small bob of mass m . The pendulum is made to oscillate with small angular amplitude θ_m . The maximum kinetic energy of the bob during the oscillation is:

- (A) $\frac{1}{4}mgL\theta_m^2$ (B) $\frac{1}{2}mgL\theta_m^2$ (C) $\frac{3}{4}mgL\theta_m^2$ (D) $mgL\theta_m^2$

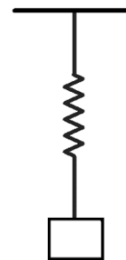
10. A thin, uniform rod of length L is pivoted at a point O at a distance $\frac{L}{4}$ from one of its ends, such that it can rotate freely in the vertical plane. From the equilibrium position shown in the figure, the rod is rotated by a small angle and released from rest. The rod performs angular SHM of time period:



- (A) $2\pi\sqrt{\frac{7L}{12g}}$ (B) $2\pi\sqrt{\frac{7L}{3g}}$
 (C) $2\pi\sqrt{\frac{21L}{4g}}$ (D) $2\pi\sqrt{\frac{21L}{g}}$

11. The specific heat at constant volume (in J/kgK) of Argon gas (molar mass = 40 g/mol) is close to: ($R = 8.31$ J/mol K)
- (A) 276 (B) 290 (C) 312 (D) 409

12. A small block of mass 0.3 kg is suspended with a vertical spring of spring constant 120 N/m as shown and released from rest at $t = 0$ with the spring in its natural length. If the vertically upward direction is taken as positive, the velocity (in cm/s) of the block varies with time (in seconds) as: ($g = 10 \text{ m/s}^2$)



- (A) $v(t) = 100 \sin\left(20t - \frac{\pi}{2}\right)$ (B) $v(t) = 100 \sin\left(20t + \frac{\pi}{2}\right)$
 (C) $v(t) = 50 \cos\left(20t - \frac{\pi}{2}\right)$ (D) $v(t) = 50 \cos\left(20t + \frac{\pi}{2}\right)$

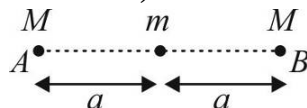
13. Let the latent heat of fusion of ice be denoted by L , and let the specific heat of water be denoted by C . A mass m_1 of ice at 0°C is mixed with mass m_2 of liquid water at $T^\circ\text{C}$, and all the ice melts. Assuming that no heat is gained from or lost to the surroundings, the final temperature is:

- (A) $\frac{m_1\left(\frac{L}{C}\right) - m_2T}{(m_1 + m_2)}$ (B) $\frac{m_2T - m_1\left(\frac{L}{C}\right)}{(m_1 - m_2)}$ (C) $\frac{m_2T - m_1\left(\frac{L}{C}\right)}{(m_1 + m_2)}$ (D) $\frac{m_2T + m_1\left(\frac{L}{C}\right)}{(m_1 - m_2)}$

14. In which of the following thermodynamic processes, the work done by the gas is positive and the change in its internal energy is negative?

- (A) Expansion at constant pressure (B) Compression at constant pressure
 (C) Adiabatic expansion (D) Adiabatic compression

15. Two particles A and B each of mass M are fixed in free space with a distance $2a$ between them. A third particle of mass m is in equilibrium at the mid-point between the particles A and B . If the particle of mass m is displaced by a small distance along the perpendicular bisector of AB , and released, it begins SHM of frequency: (G is the gravitational constant)



- (A) $\frac{1}{2\pi} \sqrt{\frac{2GM}{a^3}}$ (B) $\frac{1}{2\pi} \sqrt{\frac{GM}{a^3}}$ (C) $\frac{1}{2\pi} \sqrt{\frac{2GM}{3a^3}}$ (D) $\frac{1}{2\pi} \sqrt{\frac{GM}{2a^3}}$

16. The density of a metal at temperature T_1 is ρ_1 and its density at temperature T_2 is ρ_2 . The volume thermal expansion coefficient of the metal is given by:

- (A) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (B) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$
 (C) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (D) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$

17. A diatomic gas initially at pressure P_1 expands such that its pressure and volume follow the relation: $\frac{P}{V^3} = \text{constant}$. If volume of the gas increases from V_1 to V_2 during the process, the heat absorbed by the gas is:

- (A) $\left(\frac{7}{4}\right) \left(\frac{P_1}{V_1^3}\right) (V_2^4 - V_1^4)$ (B) $\left(\frac{11}{4}\right) \left(\frac{P_1}{V_1^3}\right) (V_2^4 - V_1^4)$
 (C) $\left(\frac{7}{4}\right) \left(\frac{P_1}{V_1^2}\right) (V_2^3 - V_1^3)$ (D) $\left(\frac{11}{4}\right) \left(\frac{P_1}{V_1^2}\right) (V_2^3 - V_1^3)$

18. A sphere made of steel is initially at the surface of a lake of depth 300 m, where the ambient temperature is 20°C . If the sphere is dropped into the lake such that it sinks and settles at the bottom, where the temperature is 10°C , the fractional change in its volume is:
(Bulk modulus of steel = 150 GPa, Volume thermal expansion coefficient of steel = $3 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$)
(Assume that the density of water everywhere in the lake is 1000 kg/m^3 , and $g = 10 \text{ m/s}^2$)
(A) 10^{-5} (B) $-(10^{-5})$ (C) $-(3 \times 10^{-5})$ (D) $-(5 \times 10^{-5})$
19. A body at temperature 80.0°C is placed in a room maintained at a constant temperature 20.0°C . The body takes time t_1 to cool from 80.0°C to 79.9°C , and it takes time t_2 to cool from 40.0°C to 39.9°C . Assuming that the cooling process follows Newton's law of cooling, the ratio $\frac{t_1}{t_2}$ is close to:
(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$
20. 100 g of Nitrogen gas is compressed at constant pressure, such that the rms speed of its molecules before and after the process is 500 m/s and 700 m/s respectively. The heat rejected by the gas during the process (in kJ) is:
(A) 12 (B) 20 (C) 24 (D) 28

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled. (Example: 6, 81, 1.50, 3.25, 0.08)

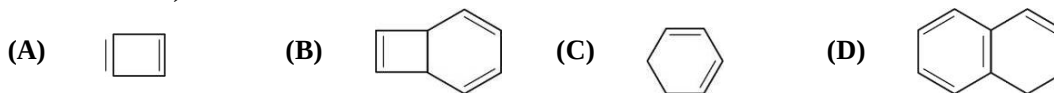
21. When 10 g of a liquid X at temperature 80°C is mixed with 20 g of a liquid Y at 20°C in a perfectly insulated vessel, the final temperature is 32°C . Now, if 5 g more of liquid Y at 20°C is added to the mixture, the final temperature (in $^{\circ}\text{C}$) is _____.
22. A gas is taken through a Carnot cycle operating between heat reservoirs at temperature 300 K and 500 K. If the heat absorbed by the gas per cycle from the heat reservoir at temperature 500 K is 20 kJ, the work done by the gas per cycle (in kJ) is _____.
23. A particle executes SHM on the Y-axis with its mean position at the origin. The speed of the particle when it is at $y = -1 \text{ cm}$ is 24 cm/s, and its speed when it is at $y = 5 \text{ cm}$ is $12\sqrt{2} \text{ cm/s}$. The maximum acceleration of the particle during the SHM (in cm/s^2) is _____.
24. An otherwise thermally insulated room can only exchange heat with the surroundings through a window with a glass pane of area 1.2 m^2 and thickness 0.5 cm. A heater placed inside the room supplies heat to the room at a constant rate 1800 W. On a day when the temperature outside the room is 14°C , the temperature inside the room (in $^{\circ}\text{C}$) a long time after the heater is switched on becomes _____. (Thermal conductivity of glass = $1 \text{ W/m }^{\circ}\text{C}$)
25. A stationary vessel contains a mixture of equal moles of Nitrogen and Helium gas. The percentage of the total kinetic energy of the mixture that is possessed by the molecules of Nitrogen is _____.

SPACE FOR ROUGH WORK

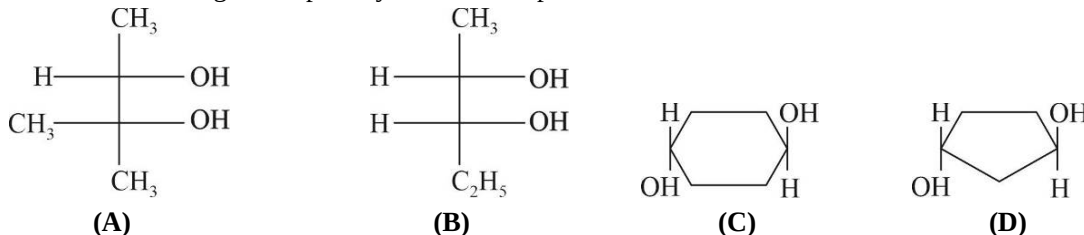
PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Which of the following compound shows complete cyclic delocalisation of $(4n + 2)\pi e^-$? (Here n is a whole number)



2. Which of following is an optically inactive compound ?



3. Given that K_{a1} and K_{a2} of H_2S at $25^\circ C$ are 10^{-8} and 10^{-12} respectively. What would be concentration of the sulphide ions in the saturated solution of $H_2S(0.1M)$ in presence of $0.1 M HCl$.



4. Identify correct order of pH of the given solutions at $25^\circ C$

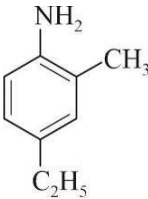


5. Z_2 undergo disproportion in an alkaline medium into a mixture of Z^- and ZO_3^- . Equivalent mass of Z_2 in the reaction is: [given that atomic mass of Z is 80 gm/mol]



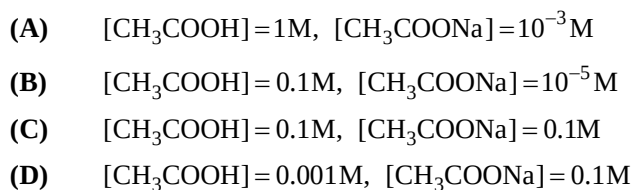
6. Given that K_{sp} of $AgCl$ is 10^{-10} and formation constant of the complex $Ag(NH_3)_2^+$ is 10^6 at $25^\circ C$. Then equilibrium constant of the reaction $AgCl(s) + 2NH_3(aq) \rightleftharpoons Ag(NH_3)_2^+ + Cl^-$ is:

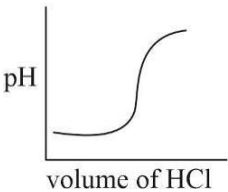


7. Correct I.U.P.A.C name of the compound  is:

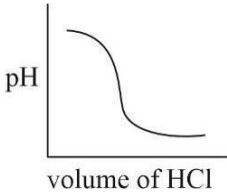


8. Which of following aqueous mixture of CH_3COOH and CH_3COONa will have maximum buffer capacity?

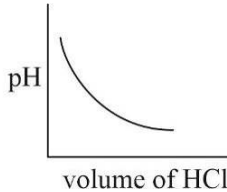


9. AgOH solution was neutralized using HNO_3 solution. If pH of the solution at 50% neutralization point is 8 at 25°C , then pK_b of AgOH is:
 (A) 4.00 (B) 5.52 (C) 6.00 (D) 9.48
10. Which of following species have maximum number of peroxide linkages per molecule?
 (A) CrO_5 (B) $\text{Cr}_2\text{O}_7^{2-}$ (C) $\text{S}_2\text{O}_8^{2-}$ (D) H_2SO_5
11. Volume strength of a H_2O_2 solution is 2.8. Normality of the solution is:
 (A) 0.1 (B) 0.5 (C) 0.28 (D) 0.56
12. An aqueous solution having 0.01 mole FeSO_4 and 0.01 mole H_2SO_4 is just sufficient to decolourise 10 ml of aqueous KMnO_4 . Then molarity of the KMnO_4 in the original solution was:
 (A) 0.40 M (B) 0.20 M (C) 0.25 M (D) 0.32 M
13. Select the increasing order of oxidation state of the oxygen atoms in following compounds.
 (I) O_2F_2 (II) OF_2 (III) H_2O_2
 (A) $\text{I} < \text{II} < \text{III}$ (B) $\text{III} < \text{II} < \text{I}$
 (C) $\text{III} < \text{I} < \text{II}$ (D) $\text{II} < \text{III} < \text{I}$
14. Which of the following diagram correctly represent variation of pH of a solution of NH_4OH by addition of increasing quantity (volume in ml) of a known HCl solution?
- 

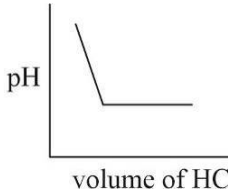
(A)

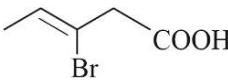


(B)

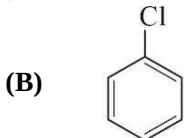


(C)

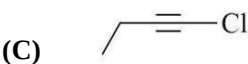


(D)
15. pH of $\frac{1}{40}$ M $\text{CH}_3\text{COONH}_4$ is: (given that K_a of CH_3COOH is 10^{-5} and K_b of NH_4OH is 10^{-5} at 25°C)
 (A) 7 (B) 6 (C) 5 (D) 8
16. Which of the following is not a redox reaction?
 (A) Reaction taking place during photosynthesis
 (B) Decomposition of H_2O_2
 (C) Decomposition of CaCO_3
 (D) Decomposition of AgBr in presence of light
17. I.U.P.A.C name of the following compound is:
- 
- (A) 3-Bromo-pent-3-en-oic acid (B) 3-Bromo-pent-2-en-oic acid
 (C) 3-Bromo-pent-4-en-oic acid (D) 3-Bromo-pentenoic acid
18. Which of following compound have weakest C – Cl bond?
- 

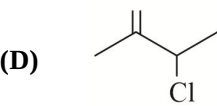
(A)



(B)



(C)



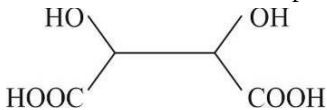
(D)

19. Select correct statement out of the following:
 (A) Eclipsed conformer of 1, 2-dichloro ethane is more stable than its Anti conformer.
 (B) Gauche conformer of ethane-1, 2-diol is more stable than its anti-conformer
 (C) Twisted boat form of cyclohexane is less stable than its Boat form
 (D) Chair form of cyclohexane is optically active
20. What would be solubility (in mole L^{-1}) of $AgCl$ ($K_{sp} = 10^{-10}$) in presence of $10^{-1} M$ $AgNO_3$?
 (A) 10^{-5} (B) 0.57×10^{-5} (C) 10^{-9} (D) 0.83×10^{-5}

SPACE FOR ROUGH WORK

SECTION-2

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21. $n_1Cr_2O_7^{-2} + n_2FeC_2O_4 \xrightarrow[\text{Medium}]{\text{Acidic}} n_3Cr^{+3} + n_4Fe^{+3} + n_5CO_2 + n_6H_2O$ here n_1, n_2, n_3, n_4, n_5 and n_6 are simplest non fractional coefficients of the balanced redox reaction. Then value of $[n_1 + n_2 + n_3 + n_4 + n_5 + n_6]$ is:
22. Number of stereoisomers possible for the following compound is:

23. 10 ml of $\frac{M}{5} H_3PO_2$ was neutralized by using $\frac{M}{5} KOH$ solution. If pK_a of H_3PO_2 is 2.0 at $25^\circ C$ then pH of the solution at the end point is:
24. Increasing amount of dilute HCl was added to an aqueous solution containing $10^{-4} M$ each of Pb^{+2} and Hg_2^{+2} . Given that K_{sp} values of $PbCl_2$ and Hg_2Cl_2 are 10^{-8} and 10^{-16} respectively at $25^\circ C$ then value of pH of the solution at which precipitation of the least soluble metal halide will start is:
25. Nitrate ion acts as an oxidising agent and gets converted into a gas of vapour density 8.5. The equivalent mass of nitrate ion is _____. (Given: Atomic mass $H = 1, N = 14, O = 16$)

SPACE FOR ROUGH WORK

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If the sum of all the solutions of the equation $2^{\cos 2x} = 3.2^{\cos^2 x} - 4$ which are lying in the interval $[0, 315]$ is $k\pi$, then the value of k is :
 (A) 1950 (B) 5050 (C) 5151 (D) 5055
- In an arithmetic progression of which a is the first term. If the sum of the first ten terms is zero and the sum of next ten terms is -200 , then the value of a is:
 (A) 7 (B) 9 (C) 11 (D) 13
- Let $45^\circ \leq \theta < 90^\circ$. If $\tan \theta + \cot \theta = \tan^2 \theta + \cot^2 \theta$ then the value of $(\sin \theta + \cos \theta)$, is :
 (A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{\sqrt{3}+1}{2}$ (D) $\frac{2}{\sqrt{3}+1}$
- The value of $\sum_{r=1}^{10} r^2 \left(\frac{{}^{26}C_r}{{}^{26}C_{r-1}} \right)$ is equal to :
 (A) 900 (B) 1100 (C) 1300 (D) 1800
- How many six-digit numbers are there in which no digit is repeated, even digits appear at even places, odd digits appear at odd places and the number is divisible by 4?
 (A) 1440 (B) 2160 (C) 2700 (D) 3600
- If lines $x - y + 2 = 0$ and $2x - y - 2 = 0$ meet at a point P then equation of tangent drawn to the parabola $y^2 = 8x$ from the point 'P' is :
 (A) $x - 2y + 8 = 0$ (B) $x + y - 16 = 0$
 (C) $3x - y - 16 = 0$ (D) $x - 3y + 16 = 0$
- The sum of rational terms in binomial expansion of $(\sqrt{2} + \sqrt[5]{3})^{10}$ is :
 (A) 32 (B) 25 (C) 41 (D) 9
- A point on the ellipse $4x^2 + 9y^2 = 36$, where the normal is parallel to the line $4x - 2y - 5 = 0$.
 (A) $\left(\frac{8}{5}, \frac{-9}{5}\right)$ (B) $\left(\frac{-9}{5}, \frac{8}{5}\right)$ (C) $\left(\frac{9}{5}, \frac{8}{5}\right)$ (D) $\left(\frac{8}{5}, \frac{9}{5}\right)$
- The area of the rectangle formed by the perpendicular from centre of $\frac{x^2}{9} + \frac{y^2}{4} = 1$, to the tangent and normal at the point whose eccentric angle $\frac{\pi}{4}$ equals :
 (A) $\frac{30}{13}$ (B) $\frac{30}{11}$ (C) $\frac{13}{30}$ (D) $\frac{26}{11}$
- On the sides AB, BC, CA of a ΔABC , 3, 4, 5 distinct points (excluding vertices A, B, C) are respectively chosen. The number of triangles that can be constructed using these chosen points as vertices are :
 (A) 220 (B) 215 (C) 210 (D) 205

11. The equation of line passing through the points of intersection of parabola $x^2 = 8y$ and ellipse $\frac{x^2}{3} + y^2 = 1$, is :
 (A) $3y + 1 = 0$ (B) $y - 2 = 0$ (C) $y + 3 = 0$ (D) $3y - 1 = 0$
12. Let the equations of two ellipse be $E_1: \frac{x^2}{3} + \frac{y^2}{2} = 1$ and $E_2: \frac{x^2}{16} + \frac{y^2}{b^2} = 1$. If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 is :
 (A) 9 (B) 16 (C) 4 (D) 2
13. If the sum of the series $25 + 24\frac{2}{5} + 23\frac{4}{5} + 23\frac{1}{5} + \dots$ is maximum, then
 (A) last term of the series is 0 (B) last term of the series is $\frac{1}{5}$
 (C) number of terms of the series are 42 (D) number of terms of the series are 40
14. The greatest value of the term independent of x in the expansion of $\left(x \sin \theta + \frac{\cos \theta}{x}\right)^{10}$ is :
 (A) $^{10}C_5$ (B) 2^5 (C) $2^5 \cdot ^{10}C_5$ (D) $\frac{^{10}C_5}{2^5}$
15. Consider the word 'CARCASSONNE'. Words are formed using all the letters of this word. If number of words which contain the word 'CAR' and vowels are separated is $k(5!)$ then k is equal to :
 (A) 157 (B) 315 (C) 630 (D) None of these
16. The sum of the infinite series $1 + \left(1 + \frac{1}{5}\right)\left(\frac{1}{2}\right) + \left(1 + \frac{1}{5} + \frac{1}{5^2}\right)\left(\frac{1}{2^2}\right) + \dots$
 (A) $\frac{20}{9}$ (B) $\frac{10}{9}$ (C) $\frac{5}{9}$ (D) $\frac{5}{3}$
17. If a, b, c are non-zero real numbers then the minimum value of the expression $\left(\frac{(a^8 + 4a^4 + 1)(b^4 + 3b^2 + 1)(c^2 + 2c + 2)}{a^4 b^2}\right)$ equals :
 (A) 24 (B) 30 (C) 48 (D) 12
18. The expression $\frac{\sin 7\theta + 6\sin 5\theta + 17\sin 3\theta + 12\sin \theta}{\sin 6\theta + 5\sin 4\theta + 12\sin 2\theta}$ when $\theta = \frac{\pi}{5}$ reduces to :
 (A) $\cos \frac{\pi}{5}$ (B) $2\sin \frac{3\pi}{10}$ (C) $\sin \frac{\pi}{5}$ (D) $2\sin \frac{\pi}{10}$
19. The focal chord of the parabola $(y - 2)^2 = 16(x - 1)$ is a tangent to the circle $x^2 + y^2 - 14x - 4y + 51 = 0$, then slope of the focal chord can be :
 (A) 0 (B) 1 (C) 2 (D) 3
20. The equation of latus rectum of a parabola is $x + y = 8$ and equation of the tangent at the vertex is $x + y = 12$, then the length of latus rectum is :
 (A) $4\sqrt{2}$ (B) $2\sqrt{2}$ (C) 8 (D) $8\sqrt{2}$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

21. If the coefficient of first three terms in the expansion $\left(\sqrt{x} + \frac{1}{2\sqrt[4]{x}}\right)^n$ where $n \in N$ form an arithmetic progression, then find the number of terms in the expansion having integral powers of x .
22. The sum of the first 10 terms of the series $\frac{7}{2^2 5^2} + \frac{13}{5^2 8^2} + \frac{19}{8^2 11^2} + \dots$ is $\frac{m}{n}$ the find the value of $(n - 12m)$?
23. A tangent is drawn to the parabola $y^2 = 48x$ which makes an angle of 30° with the axis of x . Find the distance between the tangent and a parallel normal.
24. A firm of Chartered Accountants in Bombay has to send 10 clerks to 5 different companies, two clerks in each. Two of the companies are in Bombay and the others are outside. Two of the clerks prefer to work in Bombay while three others prefer to work outside. In how many ways can the assignment be made if the preferences are to be satisfied?
25. If the straight line $4ax + 3by = 24$ is a normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$), then $(a^2 - b^2)$ is equal to_____.

SPACE FOR ROUGH WORK